Periodicity theorem

E vector bundle over the space X, and if $E_0 = E - X$, X is considered to lie in E as the zero section. The group $\mathbb{C}^*$ acts on E_0 as the group of fiber preserving automorphisms. Let $P(E) = E_0/\sim$, this is projective bundle associated to E.

Let $p : P(E) \to X$ be the projection map. So $p^{-1}(x)$ is a $\mathbb{C}$-vector space for each x. If V is a vector space and W is a one dimensional vector space, then $V \cong V \otimes W$ $v \mapsto v \otimes w$, but not naturally. This defines an isomorphism, $P(w) : P(V) \to P(V \otimes W)$. For any other nonzero element $w' \in W$, we have $w' = \lambda w$. So $P(w) = P(w')$ and the isomorphism between $P(V) \cong P(V \otimes W)$ is natural. Thus for E vector bundle and L line bundle, we have

$$P(E) \cong P(E \otimes L)$$

If E is a vector bundle over X, then each point $a \in P(E)_x = P(E_x)$ represents a one dimensional subspace $H_x^* \subset E_x$. Then

$$H^* = \bigcup_{x \in X} H_x^* \subset p^* E$$

is a subbundle, where $p : P(E) \to X$ is the projection map. Since the problem is local we may assume $E \cong X \times V$ and we have $p(E)_x = \{x\} \times Gr_1 V$.

Theorem 0.1 (Periodicity theorem)**.** *Let L be a line bundle over X. Then as a $K(X)$ algebra,*

$$K(P(L \oplus 1)) = \frac{K(X)[H]}{([H] - [1])([L][H] - [1])}$$

Note that $P(1 \oplus 1) = X \times S^2$.

Corollary 0.2.

$$K(S^2) = \frac{K(\text{pt.})[H]}{([H] - 1)^2} = \frac{\mathbb{Z}[H]}{([H] - 1)^2}$$

Corollary 0.3. *If X is any space and if*

$$\begin{aligned} \mu : K(X) \otimes K(S^2) &\xrightarrow{\simeq} K(X \times S^2) \\ a \otimes b &\mapsto \pi_1^*(a) \otimes \pi_2^*(b) \end{aligned}$$

where π_i are the projections onto the two factors, is an isomorphism of rings.

For any $x \in X$ there is a natural embedding

$$\begin{aligned} L_x &\to P(L \oplus 1)_x \\ y &\mapsto (y, 1) \end{aligned}$$

This map extends to the one point compactification of L_x onto $P(L \oplus 1)_x$

$$\begin{aligned} L_x^+ &\xrightarrow{\cong} P(L \oplus 1)_x \\ \{\infty\} &\mapsto (1, 0) \end{aligned}$$

If we map

$$\begin{aligned} s_\infty : X &\to P(L \oplus 1) \\ x &\mapsto (1,0) \end{aligned}$$

we obtain a section of $P(L \oplus 1)$, called the section of infinity.

Choose a metric for L. Let $S \subset L$ be the S^1 bundle.

$$P^0 = \{v \mid |v| \leq 1, v \in L\} \quad P^\infty = s_\infty(X) \bigcup \{v \mid |v| > 1, v \in L\}$$

We denote the projections as

$$\begin{tikzcd} S \arrow[d, "\pi"] & P^0 \arrow[d, "\pi_0"] & P^\infty \arrow[d, "\pi_\infty"] \\ X & X & X \end{tikzcd}$$

Since π_0 and π_∞ are homotopy equivalences, every bundle on P^0 is of the form $\pi_0^* E^0$ and every bundle on P^∞ is of the form $\pi_\infty^* E^\infty$, where E^0 and E^∞ are bundles on X. Thus, any bundle E on $P(L \oplus 1)$ is isomorphic to one of the form $\pi_0^*(E^0) \bigsqcup_f \pi_\infty^*(E^\infty)$ where $f \in \mathrm{Iso}(\pi^*(E^0), \pi^*(E^\infty))$ is a clutching function.

We look to approximate any gluing function using Laurent clutching functions,

$$f = \sum_{k=-n}^{k=n} a_k z^k$$

These finite series can be used to approximate functions on S in a uniform manner.

If L is trivial, then $S = S \times S^1$. and the projection $z : S \to S^1$ is just the projection map.
If $L = 1$, we have $P = \mathbb{C}P^1 \times X$ and given bundles E_0 and E_∞ over X, any choice of a section $a : X \to \mathrm{Iso}(E_\infty, E_0)$ gives a gluing function

$$S = S^1 \times X \ni (z, x) \xmapsto{f} a(z) \in \mathrm{Iso}(\pi^*(E_\infty), \pi^*(E_0))_{(z,x)}$$

To carry over the definition to a general line bundle $L \to X$ we define z as the section

$$S \ni [1 : z] \mapsto ([1 : z], z) \in \pi^* L$$

We want the gluing function f to be a section of the bundle $\mathrm{Iso}(\pi^*(E_\infty), \pi^*(E_0))$.We take coefficients a to be a section $X \to \mathrm{Iso}(E_\infty, L \otimes E_0)$ so that the formula

$$S \ni [1 : z] \xmapsto{f} f z^{-1} a(\pi([1 : z])) \in \mathrm{Iso}(\pi^*(E_\infty), \pi^* L^{-1} \otimes \pi^*(L \otimes E_0)) = \mathrm{Iso}(\pi^*(E_\infty), \pi^*(E_0))$$

makes sense.

We write L^{-k} for $\pi^*(L^k)$ and we have $L^k \otimes L^{k'} \cong L^{k+k'}$ $\forall k, k'$.

We let $L \to X$ be an arbitrary line bundle and choose $E_0 = 1$ and $E_\infty = L$ and

$$z^{-1} \in \Gamma(\pi^*(L^{-1})) = \mathrm{Iso}(1, \pi^* L^{-1}) = \mathrm{Iso}(\pi^* L, 1) = \mathrm{Iso}(\pi^* E_\infty, \pi^*(E_0))$$

a suitable gluing function. The resulting line bundle $\mathcal{V}(L, z^{-1}, 1) \simeq H$ is the canonical bundle over P.

More generally for arbitrary powers of z and index $k \in \mathbb{Z}$, a_k, we have the setup:
Given an exponent $k \in \mathbb{Z}$ and any section $a_k \in \Gamma\mathrm{Hom}(L^k \otimes E_\infty, E_0)$ the formula

$$S \ni [1 : z] \xrightarrow{f} z^k a_k(\pi([1 : z])) \in \mathrm{Iso}(\pi^*(L^k \otimes E_\infty), \pi^* L^k \otimes \pi^* E_0) = \mathrm{Hom}(\pi_* E_\infty, \pi^*(E_0))$$

defines a section $a_k z^k \in \Gamma\mathrm{Hom}(\pi^* E_\infty, \pi^* E_0)$ and we have the relation

$$\mathcal{V}(E_\infty, z^k a_k, E_0) \simeq H^k \otimes \mathcal{V}(L^k \otimes E_\infty, a_k, E_0)$$

Suppose that $f \in \Gamma\mathrm{Hom}(\pi^* E_\infty, \pi^* E_0)$, we define it's Fourier coefficients $a_k \in \Gamma\mathrm{Hom}(L^k \otimes E_0, E_\infty)$ by

$$a_k(x) = \frac{1}{2\pi i} \int_{S_x} f_x z_x^{-k-1} dz_x$$

where f_x, z_x denote the restrictions of f, z to S_x and dz_x is therefore a differential on S_x with coefficients in L_x. Let $S_n = \sum_{-n}^{n} a_k z^k$ and define the Cesaro means $f_n = \frac{1}{n} \sum_{0}^{n-1} S_k$. Then the Fejer's theorem on the $C^0(S^1)$ summability of Fourier series extends to vector bundles and gives

Proposition 0.4. *Let f be any clutching function for $\mathcal{V}(E_0, E_\infty)$ and let f_n be the seqeunce of Cesaro means of the Fourier series of f. Then f_n converges uniformly to f. Thus for large n, f_n is a clutching function for $\mathcal{V}(E_0, E_\infty)$ and*

$$\mathcal{V}(E_0, f, E_\infty) \cong \mathcal{V}(E_0, f_n, E_\infty)$$

We also get for large n, $0 \le t \le 1$, $|tf + (1 - tf_n)| \in \mathrm{Iso}(E_0, E_\infty)$.

We now look at gluing functions that are polynomial in z. Let E_0 and E_∞ be vector bundles over X and let for sone $n \in \mathbb{N}$ $p(z) = \sum_{k=0}^{n} a_k z^k$ be a polynomial gluing function of degree at most n. We introduce the auxiliary bundle $E_n^0 = \bigoplus_{k=1}^{n} L^k \otimes E_0$. Define a linearization of p by defining the section

$$\lim_n p \in \Gamma\mathrm{Hom}((\pi^*(\bigoplus_{k=0}^{n} L^k \otimes E_0)), \pi^*(E_\infty \oplus \bigoplus_{k=1}^{n} L^k \otimes E_0))$$

by the matrix

$$\mathrm{lin}_n\, p(z) = \begin{pmatrix} a_0 & a_1 & a_2 & \dots & a_n \\ -z & 1 & & & \\ & -z & 1 & & \\ & & \ddots & \ddots & \\ & & & -z & 1 \end{pmatrix}$$

It has properties $\mathrm{lin}_n(p \oplus p') = \mathrm{lin}_n\, p \oplus \mathrm{lin}_n\, p'$ and $\mathrm{lin}_n(\mathrm{id}_{\pi^* D} \otimes p) = \mathrm{id}_{\pi^* D} \otimes \mathrm{lin}_n\, p$ for a bundle $D \to X$.

The (i, j)th entry of the matrix $\mathrm{lin}_n\, p$ acts on jth summand and takes it to the ith summand. So lin is linear in z.

Define a sequence $p_r(z)$ inductively by: $p_0 = p, zp_{r+1}(z) = p_r(z) - p_r(0)$. Then we can write $\text{lin}_n(p) = (1+N_1)(p\oplus 1)(1+N_2)$ where N_i are nilpotent matrices. If N is nilpotent, then $1+tN$ is nonsingular for $0 \leq t \leq 1$.

Proposition 0.5. *$\text{lin}_n(p)$ and $p \oplus 1$ define isomorphic bundles on P.*

$$\mathcal{V}(E_0, p, E_\infty) \bigoplus \mathcal{V}\left(\sum_{k=1}^{n} L^k \otimes E_0, 1, \sum_{k=1}^{n} L^k \otimes E_0\right) \cong \mathcal{V}\left(\sum_{k=0}^{n} L^k \otimes E_0, \lim_{n}(p), E_\infty \oplus \sum_{k=1}^{n} L^k \otimes E_0\right)$$

Lemma 0.6. *Let p be a polynomial clutching function of degree $\leq n$ for (E_0, E_∞). Then*

1. $\text{lin}_{n+1}(E_0, p, E^\infty) \cong \text{lin}_n(E_0, p, E^\infty) \oplus \left(\text{lin}_{n+1} \otimes E_0, 1, \text{lin}_{n+1} \otimes E^0\right)$.
2. $\text{lin}_{n+1}\left(L^{-1} \otimes E_0, zp, E^\infty\right) \cong \text{lin}_n(E_0, p, E^\infty) \oplus \left(L^{-1} \otimes E_0, z, E_0\right)$

Proposition 0.7. *For any polynomial clutching fucntion p for (E_0, E_∞), we have the identity in $K(P)$*

$$([E_0, p, E_\infty] - [E_0, 1, E_0])(LH - 1) = 0$$

If we take $E_0 = 1, p = z, E_\infty = L$ we get $(H-1)(LH-1) = 0$.

Suppose $T \in \text{End}(E)$ for some finite dimensional vector space E. Let S be a circle in $\mathbb{C}$ which does not contain any eigenvalue of T. Then

$$Q_T = \frac{1}{2\pi i}\int_S \frac{1}{zI - T} dz$$

is a projection operator in E which commutes with T. The decomposition

$$E = E_+ \oplus E_-$$

where $E_+ = QE, E_- = (1-Q)E$ is invaraint under T. So T can be decomposed as

$$T = T_+ \oplus T_-$$

T_+ has all eigenvalues inside S while T_- has all of its eigenvalues outside S.

Proof. $T_\lambda I$ is invertible $\forall z \in S^1$ and commutes with T, so does Q_T. So Q_T preserves the eigenspaces of T. We can prove the proposition by considering just one eigenvalue.

$|T - \lambda I|$ is invertible iff $\lambda \notin \sigma(T)$. So if eigenvalue $|\lambda| > 1$, then $T - \lambda I$ is invertible inside D^2 and if $|\lambda| < 1$, then $T - \lambda I$ is homolomorphic outisde D^2 with a possible pole at ∞. By change of variables $z \mapsto 1/z$, we can see that $Q_T = I$. □

We defined $p(z)$ as sections over S. We extend the linear gluing function l, over the disk bundle $P_0 \to X$ to make it a section in $\Gamma(\pi_0^* E_0, \pi_0^* E_\infty)$. While similarly $1/zl(z)$ is a section in $\Gamma(\pi_\infty^* E_0, \pi_\infty^* E_\infty)$. Over $X_\infty \subset P_\infty$ the latter restricts to a. The values of these sections off S no longer need to be isomorphism. (*Atiyah*) If we assert that $l(z)$ is an isomophism outside S, we shall take this to include the statement that a is an isomorphism.

Proposition 0.8. *Let l be a linear clutching function for (E_0, E_∞) and define endomorphism Q^0, Q^∞ of E_0, E_∞ by putting*

$$Q_x^0 = \frac{1}{2\pi i}\int_{S_x} l^{-1}\mathrm{d}l \quad Q_x^\infty = \frac{1}{2\pi i}\int_{S_x} \mathrm{d}l\, l$$

Then Q^0, Q^∞ are projection operators and $pQ^0 = Q^\infty p$.

Write $E_+^i = Q^i E^i, E_-^i = \left(1 - Q^i\right) E^i, i = 0, \infty$, so that $E^i = E_+^i \oplus E_-^i$. Then p is compatible with these decompositions, so that $p = p_+ \oplus p_-$. Moreover, $p_+$is an isomorphism outside S, and $p_-$is an isomorphism inside S

Proof. WLOG we can assume $X = \mathrm{pt}$, so $L = \mathbb{C}$ and z is just a complex number. Since l is an isomorphism for $|z| - 1$, and the set of isomorphisms is an open subset of hom, we can find a real number α, with $\alpha > 1$ such that $l(\alpha) : E_0 \to E_\infty$ is an isomophism. We can now identify E_0, E_∞ with this isomorphism.

Consider the Möbius map $w = \frac{1-\alpha z}{z-\alpha}$ which preserves the unit circle and its inside. This gives $l(z) = \frac{w-T}{w+\alpha}$ where $T \in \mathrm{End}(E_0)$. We get $Q^0 = Q^\infty = Q$. We can infer from this that l_+ and l_- are isomorphisms. □

Corollary 0.9. *Let l be a linear gluing function write*

$$l_+ = a_+ z + b_+, \quad l_- = a_- z + b_-$$

Then, if $l(t) = l_+(t) \oplus l_-(t)$, where

$$l_+(t) = a_+ z + t b_+, \quad l_-(t) = t a_- z + b_-, \quad 0 \le t \le 1,$$

we obtain a homotopy of linear clutching functions connecting p with $a_+ z \oplus b_-$. Thus

$$\left(E^0, l, E^\infty\right) \cong \left(E_+^0, z, L \otimes E_+^0\right) \oplus \left(E_-^0, 1, E_-^0\right)$$

We go back to a polynomial clutching function p of degree $\le n$. Then $\mathrm{lin}_n(p)$ is a linear clutching function for (V^0, V^∞) where $V^0 = \displaystyle\sum_{k=0}^{n} L^k \otimes E^0 \quad V^\infty = E^\infty \oplus \sum_{k=1}^{n} L^k \otimes E^0$.

Hence it defines a decomposition $V^0 = V_+^0 \oplus V_-^0$. To express the dependence on p and n. we write

$$V_+^0 = \mathcal{V}_n(E_0, p, E_\infty)$$

This is a vector bundle on Z. If p_t is a homotopy of polynomial clutching functions of degree $\le n$ then

$$\mathcal{V}_n\left(E^0, p_0, E^\infty\right) \cong \mathcal{V}_n\left(E^0, p_1, E^\infty\right)$$

This gives us

$$\mathcal{V}_{n+1}\left(E^0, p, E^\infty\right) \cong \mathcal{V}_n\left(E^0, p, E^\infty\right),$$
$$\mathcal{V}_{n+1}\left(L^{-1} \otimes E^0, zp, E^\infty\right) \cong \mathcal{V}_n\left(E^0, p, E^\infty\right) \oplus \left(L^{-1} \otimes E^0\right)$$

and

$$\mathcal{V}_{n+1}\left(E^0, zp, L \otimes E^\infty\right) \cong L \otimes \mathcal{V}_n\left(E^0, p, E^\infty\right) \oplus E^0$$

Combining this with the above corollary and **??**, we obtain the following formula in $K(P)$:

$$\begin{aligned}
\left[E^0, p, E^\infty\right] + \left\{\sum_{k=1}^{n}\left[L^k \otimes E^0\right]\right\}[1] &= \left[v_n\left(E^0, p, E^\infty\right)\right]\left[H^{-1}\right] \\
+ \left\{\sum_{k=0}^{n}\left[L^k \otimes E^0\right] - \left[v_n\left(E^0, p, E^{\infty g}\right)\right]\right\}[1].
\end{aligned}$$

and hence the formula

$$\left[E^0, p, E^\infty\right] = \left[\mathcal{V}_n\left(E^0, p, E^\infty\right)\right]\left(\left[H^{-1}\right] - [1]\right) + \left[E^0\right][1].$$

This shows that $[\mathcal{V}_n\left(E^0, p, E^\infty\right)] \in K(X)$ completely determines $[E_0, p, E_\infty] \in K(P)$.

Let t be an indeterminant over the ring $K(X)$. Then the map have a $K(X)$ algebra homomorphism

$$\begin{aligned}
\mu : \frac{K(X)[t]}{(t-1)(Lt-1)} &\to K(P) \\
t &\mapsto [H]
\end{aligned}$$

To show μ is an isomorphism, we construct an inverse.

First suppose that f is a clutching function for (E_0, E_∞). let f_n be the sequence of Cesaro menas of its Fourier series, and put $p_n = z^n f_n$. Then for large n, p_n is a polynomial clutching function (of degree $\leq 2n$) for $(E_0, L^n \otimes E_\infty)$. We define

$$v_n(f) := [\mathcal{V}_{2n}(E_0, p_n, L^n \otimes E_\infty)](t^{n-1} - t^n) + [E_0]t^n$$

For large n, we have a homotopy between p_{n+1} and zp_n. This gives us

$$\mathcal{V}_{2n+2}(E_0, p_{n+1}, L^{n+1} \otimes E_\infty) \cong L \otimes \mathcal{V}_{2n}(E_0, p_n, L^n \otimes E_\infty) \oplus E_0$$

Henc $v_{n+1}(f) = v_n(f)$. Thus $v_n(f)$ is independent of n for large n. We write it as $v(f)$.

If g is close to f and n is large, then $f_n \simeq_n$ as polynomial clutching functions of degree $\leq 2n$ and hence

$$v(f) = v_n(f) = v_n(g) = v(g)$$

Thus $v(f)$ is a locally constant function of f and hence depends only on the homootpy class of f. However if E is any bundle on P and f a clutching function defining E, we define $v(E) = v(f)$. This is well-defined, additive, and depends only on isomorphism class of E. We have a group homomorphism

$$v : K(P) \to \frac{K(X)[t]}{(t-1)(Lt-1)}$$

$K(P)$ is additively generated by elements of the form $[E]$, so $\mu v(E) = E$ proves $\mu v = \mathrm{id}$. Similarly $v\mu(t^n) = t^n$ for all $n \geq 0$. So $v\mu = \mathrm{id}$.